SEND5232: Software Architecture and Design Project

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PHASE 1.2 — ARCHITECTURAL STRUCTURE & INITIAL DESIGN

📚 Lihq’et Bookstore System

A. Component Diagram (LOGICAL VIEW)

Components:

Frontend
• HTML, CSS, JavaScript (User Interface)

Backend
• Controller Layer (Handles API requests/responses)
• Service Layer (Business logic & validation)
• Model Layer (Database interaction & SQL queries)

Database
• MySQL Database (Book records storage)

Interactions:
• Frontend → Controller: Sends HTTP requests (CRUD operations)
• Controller → Service Layer: Forwards requests for processing
• Service Layer → Model: Executes business logic and database operations
• Model → Database: Performs SQL queries
• Database → Model → Service → Controller → Frontend: Returns data/response

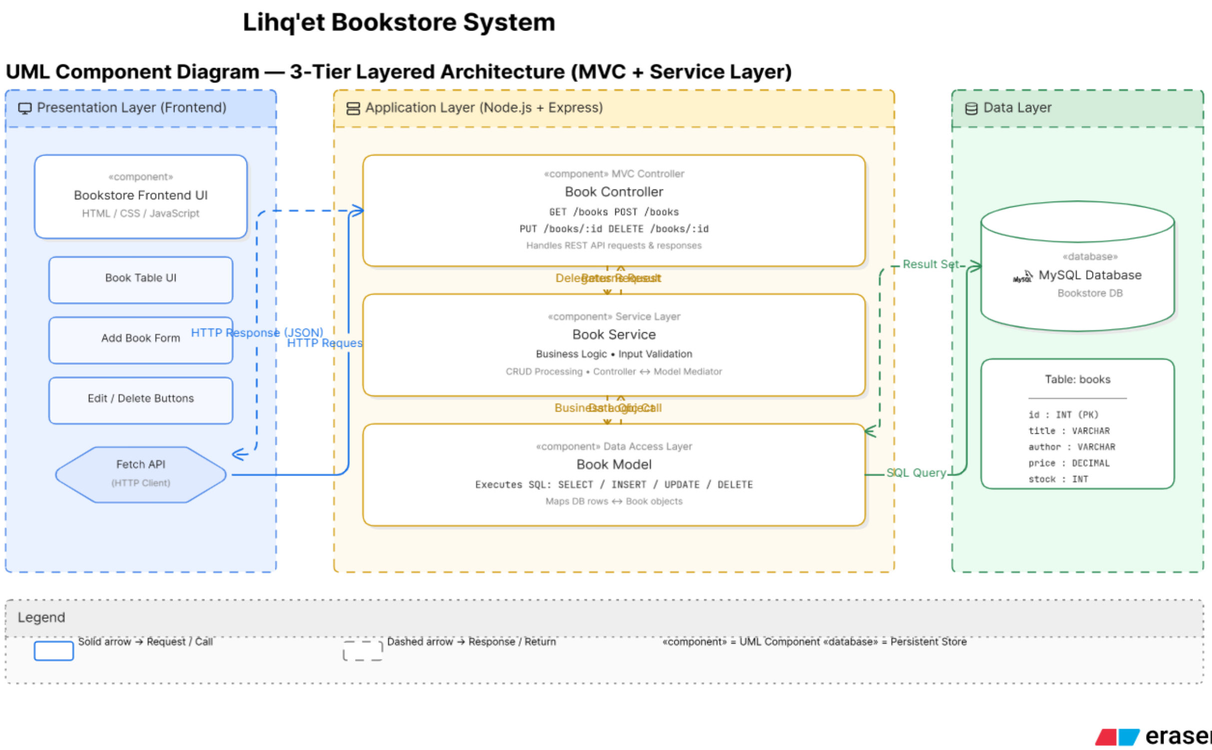

Figure 1: Component Diagram of Lihq' et Bookstore System

The component diagram presents the internal logical structure of the Lihq'et Bookstore System based on a 3-tier layered architecture combined with the MVC and Service Layer patterns. The frontend component, built using HTML, CSS, and JavaScript, is responsible for user interaction and communicates with the backend through HTTP requests. The backend is divided into three main components: the Controller, Service Layer, and Model. The Controller handles incoming API requests and forwards them to the Service Layer, which contains the business logic and validation rules. The Service Layer interacts with the Model component, which is responsible for executing SQL queries and communicating with the MySQL database. The database stores and retrieves book records. This layered separation ensures low coupling, high cohesion, and maintainability while clearly illustrating how data flows between components.